

ECE 4644 / 5610 Satellite Communications**Homework Assignment #10 – Demo - Complete****Problem #1**

Alphanumeric characters are transmitted as 7-bit ASCII words, with a single parity bit added, over a link with a transmission rate of 9.6 kbps.

- How many characters are transmitted each second?
- If a typical page of text contains 500 words with an average of five characters per word and a space between words, how long does it take to transmit a page?
- If the bit error rate on the link is 10^{-5} , how many characters per page are detected as having errors? How many undetected errors are there?
- On average, how many pages can be transmitted before (i) a detected error occurs or (ii) an undetected error occurs?

Problem #2

- Explain what MESH and STAR architectures are in a VSAT network.
- Give two advantages and disadvantages of the MESH and STAR architectures.
- Give the three major types of multiple access schemes that are used in satellite systems.
- Which one of the three multiple access schemes identified in (c) is the closest to the ALOHA multiple access scheme?
- What are the advantages and disadvantages of an MF-TDMA access scheme from a system perspective (i.e. how does this access scheme affect the earth terminal and the satellite payload design)?

Problem #3

- Describe the two attenuation bands of greatest significance to satellite communications.

Rain attenuation for frequencies below 25 GHz can be modeled as

$$A = aR^b \text{ (dB/km)}$$

where R is the rain rate in mm/hr and a and b are factors related to frequency, f , in GHz, according to

$$a = 4.21 \times 10^{-5} f^{2.42}$$

$$b = 1.41 \times f^{-0.078}$$

- b) For the situation of stratiform rain with $R = 10$ mm/hr that extends from the height of the transmitter at sea level to the melting height at 5 km along a propagation path that makes an angle of 30° with the horizon on its way to a satellite while operating on a frequency of 18 GHz, find the total attenuation of signal power due to the rain.
- c) For the situation described in part (b) does an increase in sky noise temperature on this uplink impact C/N for the uplink?

Problem #4

- a) Explain what is meant by the term collision in a random access system.
- b) Discuss the use of ARQ to correct bit errors in data packets and name three ARQ techniques.
- c) What is the probability of a 1-bit error occurring in a 5-bit word with $BER = 10^{-3}$? This is basically the probability of a word error occurring (PE).
- d) Suppose a single parity bit is added to the 5-bit word to make up a 6-bit codeword. Now 1-bit errors are detected and corrected with ARQ. Calculate the resulting PE for the codeword.
- e) Determine the BER for a 5-bit word that corresponds to this PE (may require iteration). Find the effective coding gain obtained by implementing the 1-bit parity checking.
- f) Explain whether this is an example of FEC and describe the cost to the link of implementing this form of error correction.

ECE 4644 / 5610 Satellite Communications**Homework Assignment #10 – Demo - Solutions****Problem #1**

Alphanumeric characters are transmitted as 7-bit ASCII words, with a single parity bit added, over a link with a transmission rate of 9.6 kbps.

- a) How many characters are transmitted each second?

The codeword for each character consists of a 7-bit message sequence followed by a single parity bit, for a total length of 8 bits. The number of characters transmitted per second is $9.6 \times 10^3 / 8 = 1200$ characters per second.

- b) If a typical page of text contains 500 words with an average of five characters per word and a space between words, how long does it take to transmit a page?

The transmission of an average word requires five characters plus one character for a space for a total of six characters. Then the total number of characters required to transmit a single page of text is $500 \times 6 = 3000$ characters. The time to transmit one page is $3000 \text{ characters} / 1200 \text{ characters per second} = 2.5 \text{ sec}$.

- c) If the bit error rate on the link is 10^{-5} , how many characters per page are detected as having errors? How many undetected errors are there?

The 1-bit parity checking will detect errors when 1, 3, 5, or 7 bits are in error. The rate at which errors are detected will be dominated by the 1-bit error rate. This is found from the binomial distribution $P(k) = \binom{n}{k} p^k (1-p)^{n-k}$ with $n = 8$ and $k = 1$:

$$P(1) = \binom{8}{1} 10^{-5} (1 - 10^{-5})^{8-1} = \frac{8!}{1!7!} 10^{-5} (1 - 10^{-5})^7 = 8 \times 10^{-5}$$

And the number of detected errors per page is $8 \times 10^{-5} \times 3000 = 0.24 \text{ errors/page}$.

The 1-bit parity checking will not detect errors when 2, 4, 6, or 8 bits are in error. The rate at which errors are not detected will be dominated by the 2-bit error rate. This is found from the binomial distribution $P(k) = \binom{n}{k} p^k (1-p)^{n-k}$ with $n = 8$ and $k = 2$:

$$P(2) = \binom{8}{2} (10^{-5})^2 = \frac{8!}{2!6!} (10^{-5})^2 (1 - 10^{-5})^6 = 2.8 \times 10^{-9}$$

And the number of undetected errors per page is $2.8 \times 10^{-9} \times 3000 = 8.4 \times 10^{-6} \text{ errors/page}$.

- d) On average, how many pages can be transmitted before (i) a detected error occurs or (ii) an undetected error occurs?

For (i) a detected error occurs on average every $1 / 0.24 = 4.2 \text{ pages}$ while for (ii) an undetected error occurs on average every $1 / 8.4 \times 10^{-6} = 119,000 \text{ pages}$.

Problem #2

- a) Explain what MESH and STAR architectures are in a VSAT network.

A MESH VSAT architecture allows each of the VSATs in the network to communicate directly with any of the other VSATs through the satellite. A STAR VSAT architecture only allows the VSATs to communicate with each other via the hub earth station.

- b) Give two advantages and disadvantages of the MESH and STAR architectures.

Two MESH network advantages: direct communication between VSATs; no double hops between VSATs. Two MESH disadvantages: low data rates in both direction (no large hub to compensate for small antennas at each end of the loop); low satellite transponder utilization leading to high space segment costs. Two STAR network advantages: relatively high data rates overall; high space segment utilization and thus lower space segment costs. Two STAR disadvantages: double hops required between VSATs; VSATs unable to communicate directly with each other.

- c) Give the three major types of multiple access schemes that are used in satellite systems.

The three major types of multiple access schemes used in satellite systems are: FDMA, TDMA, and CDMA.

- d) Which one of the three multiple access schemes identified in (c) is the closest to the ALOHA multiple access scheme?

ALOHA is closest to TDMA in its design, particularly slotted (constrained) ALOHA for which transmissions occur only within designated time intervals.

- e) What are the advantages and disadvantages of an MF-TDMA access scheme from a system perspective (i.e. how does this access scheme affect the earth terminal and the satellite payload design)?

MF-TDMA is a combination of FDMA (which is the lowest cost and most flexible of all the access schemes) and TDMA (which is the most spectrally efficient access scheme in a non-interference limited environment). By using a relatively narrow band for the uplink, the terminal costs are kept reasonably low, and by having a narrow band TDMA scheme within the FDMA allocation, there is an added flexibility to the system by allowing a number of terminals to access the satellite simultaneously. The disadvantage of an MF-TDMA scheme is mainly due to the generic TDMA requirement to have the burst rate of every terminal much higher than the individual stream from the terminal would require on its own. There are also synchronization issues as with all TDMA schemes and the need to have timing control closely monitored.

Problem #3

a) Describe the two attenuation bands of greatest significance to satellite communications.

In terms of rising frequency in the GHz range, the first significant attenuation peak occurs near 22 GHz. It is due to resonant absorption by water vapor molecules. The net attenuation on a satellite link amounts to 1-2 dB. This is sufficient for the 'passbands' at lower (Ku) and higher (Ka) frequencies to be preferred.

The next attenuation peak occurs near 60 GHz and is due to resonant absorption by molecular oxygen. The rate of attenuation is so high, > 10 dB/km, that this frequency is unusable for long-range propagation in the atmosphere. However, it is a good choice for supporting radio links between satellites above the atmosphere.

Rain attenuation for frequencies below 25 GHz can be modeled as

$$A = aR^b \text{ (dB/km)}$$

where R is the rain rate in mm/hr and a and b are factors related to frequency, f , in GHz, according to

$$a = 4.21 \times 10^{-5} f^{2.42}$$

$$b = 1.41 \times f^{-0.078}$$

b) For the situation of stratiform rain with $R = 10$ mm/hr that extends from the height of the transmitter at sea level to the melting height at 5 km along a propagation path that makes an angle of 30° with the horizon on its way to a satellite while operating on a frequency of 18 GHz, find the total attenuation of signal power due to the rain.

Solve for the factors by direct substitution with $f = 18$ [GHz]:

$$a = 0.0460$$

$$b = 1.125$$

$$\text{Then } A = (0.0460) 10^{1.125} = 0.61 \text{ dB/km}$$

The length of the propagation path through the rain is given by $5 \text{ km} / \sin(30^\circ) = 10 \text{ km}$.

Thus, the net attenuation is about 6 dB.

c) For the situation described in part (b) does an increase in sky noise temperature on this uplink impact C/N for the uplink?

The rain on an uplink does not change the sky noise temperature for the satellite because the earth-directed receive antenna always has the 'warm' Earth filling the beam. If the rain had occurred on the downlink, we would expect an increase in sky noise temperature and hence in overall noise power, leading to an additional reduction in C/N.

Problem #4

- a) Explain what is meant by the term *collision* in a random access system.

In a random access (RA) system packets from multiple sources are routed through a single satellite transponder channel without separation by frequency, time, or coding. If packets from two or more sources coincide in time the signals are scrambled and their information lost. The interference of the packets in the transponder is called a collision. Packets lost to collisions may be recovered by ARQ techniques.

- b) Discuss the use of ARQ to correct bit errors in data packets and name three ARQ techniques.

ARQ refers to Automatic Repeat Request. The receiving end of a satellite link inspects a received packet for signs of corruption (bit errors). If the packet is deemed to be good, an acknowledgement (ACK) may be sent to the transmitting end of the link, causing the next packet to be transmitted. If the packet is deemed to be bad, a Negative acknowledgement (NACK) is sent and the transmitting end resends the packet. Three ARQ techniques in common usage are Stop-and-wait, Go-back-N, and Continuous transmission.

- c) What is the probability of a 1-bit error occurring in a 5-bit word with $BER = 10^{-3}$? This is basically the probability of a word error occurring (PE).

The probability of a 1-bit error is given by the binomial distribution $P(k) = \binom{n}{k} p^k (1-p)^{n-k}$ with $n = 5$ and $k = 1$. The binomial coefficient is $\binom{5}{1} = 5!/1!(5-1)! = 120/24 = 5$ and we have that $P(1) = 5 \cdot 10^{-3} (1 - 10^{-3})^4 = 0.00498 = 4.98 \times 10^{-3}$. Although 2-bit and higher errors do occur they are so much less likely that this defines the basic probability of word error and so $PE = 4.98 \times 10^{-3}$. This corresponds to one error every 200 words.

- d) Suppose a single parity bit is added to the 5-bit word to make up a 6-bit codeword. Now 1-bit errors are detected and corrected with ARQ. Calculate the resulting PE for the codeword.

Since the 1-bit errors are now corrected with ARQ the PE is determined by the occurrence of 2-bit errors within 6-bit codewords. Applying the binomial distribution again with $n = 6$ and $k = 2$ we obtain $\binom{6}{2} = 6!/2!(6-2)! = 720/48 = 15$ and we have that $P(2) = 15 (10^{-3})^2 (1 - 10^{-3})^4 = 0.000015 = 1.5 \times 10^{-5}$. The probability of word error is now $PE = 1.5 \times 10^{-5}$. This corresponds to one error every 66,700 words.

- e) Determine the BER for a 5-bit word that corresponds to this PE (may require iteration). Find the effective *coding gain* obtained by implementing the 1-bit parity checking.

We want to find the BER value that produces $PE = 1.5 \times 10^{-5}$ for a 5-bit word. That is, find p such that $P(k) = \binom{5}{1} p^1 (1-p)^{5-1} = 5p(1-p)^4 = 1.5 \times 10^{-5}$. A starting guess is obtained from $5p(1)^4 = 1.5 \times 10^{-5}$ which gives $p = 3.0 \times 10^{-6}$. Check: $P(1) = 1.5 \times 10^{-5}$. So the 5-bit BER that produces the same PE is $BER = 3 \times 10^{-6}$.

To find the coding gain from adding the parity bit, determine the C/N equivalents of the 5-bit BERs, namely, $BER = 10^{-3}$ and $BER = 3 \times 10^{-6}$. Assuming QPSK modulation, we have $P(\text{error}) = Q[z] = Q[\sqrt{C/N}]$. Examine the Q tables. For $BER = 10^{-3}$ we have $z = 3.1 = \sqrt{C/N}$ and so $C/N = 9.6$ or 9.8 dB. For $BER = 3 \times 10^{-6}$ we have $z = 4.5 = \sqrt{C/N}$ and so $C/N = 20.25$ or 13.1 dB. Therefore the coding gain from implementing the 1-bit parity checking is $13.1 - 9.8 = \underline{3.3 \text{ dB}}$.

- f) Explain whether this is an example of *FEC* and describe the cost to the link of implementing this form of error correction.

When Forward Error Correction (FEC) is applied extra bits are added to the transmissions so that errors can be detected on receive and corrected. In this example errors are detected with the 1-bit parity check and corrected with ARQ. Strictly speaking this is not FEC, because true FEC moves only forward and not back (ARQ goes back with re-send requests). With FEC the information available to correct bad words is carried with the transmission in the form of block codes, convolutional codes, etc. True FEC is easy to switch in and out because you don't need to adjust your protocol, as we had to do when changing from 5-bit words to 6-bit codewords with ARQ.

The cost to the link performance of implementing this form of error correction is a lowered rate of message transmission because of the time needed to send the extra bits. In this case the message transmission rate with 6-bit codewords is 5/6 of the rate with 5-bit words. If the message rate must be maintained, then the link bit rate must be increased by a corresponding amount, i.e., 6/5.

